

Conditions for Oscillation in the Wilson-Cowan Model

Keyue Jiang

56511325@cuni.cz

Abstract

We derive a necessary condition for oscillation in the simplified Wilson-Cowan model from the Jacobian trace and determinant: $w_{EI}w_{IE} > 16\tau_E w_{II}/(\tau_I a_E)$. The NMA default parameters violate it ($52 < 73.3$), ruling out oscillation under any external input. We tighten this bound via the I-nullcline geometry and validate over $N = 5000$ random parameter sets with zero violations.

1 Model

In computational neuroscience, the **Wilson-Cowan model** (Hugh R. Wilson & Jack D. Cowan) describes the dynamics of interactions between populations of very simple excitatory and inhibitory model neurons. It is a mean-field model that tracks the **proportion of active neurons** in each population over time.

$$\tau_E \frac{dr_E}{dt} = -r_E + F_E(w_{EE} \cdot r_E - w_{EI} \cdot r_I + I_E^{\text{ext}}) \quad (1)$$

$$\tau_I \frac{dr_I}{dt} = -r_I + F_I(w_{IE} \cdot r_E - w_{II} \cdot r_I + I_I^{\text{ext}}) \quad (2)$$

where the activation function F is a sigmoid:

$$F(x; a, \theta) = \frac{1}{1 + e^{-a(x-\theta)}} - \frac{1}{1 + e^{a\theta}} \quad (3)$$

The correction term $-\frac{1}{1+e^{a\theta}}$ ensures $F(0) = 0$ (no input yields no activation).

Remark: The original Wilson-Cowan (1972) paper derives integral equations with absolute refractoriness, then approximates them via time coarse-graining into ODEs retaining a refractory factor $(1 - r_E)$. We use the further simplified ODE formulation from Neuromatch Academy, which omits this refractory term entirely. All results in this report apply only to this simplified system.

1.1 Variables

- $r_E(t)$ — average firing rate (or proportion of active neurons) of the **excitatory** population at time t . Ranges from 0 (silent) to 1 (maximally active).
- $r_I(t)$ — average firing rate of the **inhibitory** population at time t .

1.2 Parameter Values

Parameter	Value	Description
τ_E	1.0 ms	Average synaptic time constant for excitatory population
τ_I	2.0 ms	Average synaptic time constant for inhibitory population
a_E	1.2	Excitatory gain
a_I	1.0	Inhibitory gain
θ_E	2.8	Excitatory threshold
θ_I	4.0	Inhibitory threshold
w_{EE}	9.0	Strength of self-excitatory feedback
w_{EI}	4.0	Strength of connections between E-I
w_{IE}	13.0	Strength of connections between I-E
w_{II}	11.0	Strength of self-inhibitory feedback
I_E^{ext}	0.0	External input to excitatory population (sweep this)
I_I^{ext}	0.0	External input to inhibitory population

Table 1: Default parameter values from the Neuromatch Academy computational neuroscience curriculum.

2 Nullcline Analysis

2.1 E-nullcline

Let $\dot{r}_E = 0$, then

$$\frac{1}{\tau_E}(-r_E + F_E(w_{EE} \cdot r_E - w_{EI} \cdot r_I + I_E^{\text{ext}})) = 0 \quad (4)$$

$$-r_E + F_E(w_{EE} \cdot r_E - w_{EI} \cdot r_I + I_E^{\text{ext}}) = 0 \quad (5)$$

$$F_E(w_{EE} \cdot r_E - w_{EI} \cdot r_I + I_E^{\text{ext}}) = r_E \quad (6)$$

Then we use the inverse of F (namely the logit function):

$$F^{-1}(y; a, \theta) = \theta + \frac{1}{a} \ln \frac{y + c}{1 - y - c}, \quad c = \frac{1}{1 + e^{a\theta}} \quad (7)$$

Applying F_E^{-1} to both sides:

$$F_E^{-1}(r_E) = w_{EE} \cdot r_E - w_{EI} \cdot r_I + I_E^{\text{ext}} \quad (8)$$

Solving for r_I :

$$r_I = \frac{1}{w_{EI}}(w_{EE} \cdot r_E - F_E^{-1}(r_E) + I_E^{\text{ext}}) \quad (9)$$

2.2 I-nullcline

Analogously, let $\dot{r}_I = 0$, we have

$$r_I = F_I(w_{IE} \cdot r_E - w_{II} \cdot r_I + I_I^{\text{ext}}) \quad (10)$$

$$r_E = \frac{1}{w_{IE}}(w_{II} \cdot r_I + F_I^{-1}(r_I) - I_I^{\text{ext}}) \quad (11)$$

3 Qualitative Analysis

We use the Jacobian matrix to classify the fixed points. The Jacobian is:

$$J = \begin{pmatrix} \frac{1}{\tau_E}(-1 + w_{EE} \cdot F'_E) & \frac{1}{\tau_E}(-w_{EI} \cdot F'_E) \\ \frac{1}{\tau_I}(w_{IE} \cdot F'_I) & \frac{1}{\tau_I}(-1 - w_{II} \cdot F'_I) \end{pmatrix} \quad (12)$$

in which

$$F'(x) = a \cdot (F(x) + c)(1 - F(x) - c), \quad c = \frac{1}{1 + e^{a\theta}} \quad (13)$$

Note that F' is evaluated at the sigmoid input, not at r_E or r_I directly:

$$u_E = w_{EE}r_E - w_{EI}r_I + I_E^{\text{ext}}, \quad u_I = w_{IE}r_E - w_{II}r_I + I_I^{\text{ext}} \quad (14)$$

We classify each fixed point using the trace $\tau = \text{tr}(J)$, determinant $\Delta = \det(J)$, and discriminant $\tau^2 - 4\Delta$ of the Jacobian:

- $\Delta < 0 \Rightarrow$ saddle point
- $\Delta > 0, \tau < 0 \Rightarrow$ stable (node if $\tau^2 - 4\Delta > 0$, spiral if < 0)
- $\Delta > 0, \tau > 0 \Rightarrow$ unstable

For our 3 fixed points:

Fixed Point	τ	Δ	$\tau^2 - 4\Delta$	Classification
$(0, 0)$	-1.247	0.406	-0.069	Stable Spiral
$(\approx 0.34, 0.17)$	$-$	-0.941	$-$	Saddle Point
$(\approx 0.94, 0.68)$	-2.403	1.381	0.250	Stable Node

Table 2: Fixed point classification for default parameters ($I_E^{\text{ext}} = 0$).

The system with default parameters ($I_E^{\text{ext}} = 0$) is therefore **bistable**.

Throughout this report, “oscillation” refers to sustained periodic oscillation (limit cycle), not transient damped oscillation around a stable spiral.

4 Conditions for Oscillation

The **Poincaré-Bendixson theorem** states that for a two-dimensional system, any non-empty compact ω -limit set containing only finitely many fixed points is either (1) a fixed point, (2) a periodic orbit, or (3) a connected set composed of finitely many fixed points together with homoclinic and heteroclinic orbits connecting them.

Our goal is to find the limit cycle. The system is bounded by the sigmoid, so forward orbits remain in a compact region. If this region contains a unique fixed point and that fixed point is an unstable node or spiral (both eigenvalues having positive real parts), then no trajectory other than the fixed point itself converges to it in forward time. The ω -limit set of any other trajectory therefore contains no fixed point, excluding cases (1) and (3), so it must be a periodic orbit. We now derive conditions on the structural parameters under which a unique unstable fixed point can exist.

4.1 Trace and determinant conditions

An unstable fixed point requires the Jacobian to have both a positive trace (instability) and a positive determinant (node or spiral, rather than saddle):

$$\tau = \frac{1}{\tau_E}(-1 + w_{EE}F'_E) + \frac{1}{\tau_I}(-1 - w_{II}F'_I) \quad (15)$$

$$\Delta = \frac{1}{\tau_E \tau_I} [1 + w_{II}F'_I - w_{EE}F'_E + (w_{EI}w_{IE} - w_{EE}w_{II})F'_E F'_I] \quad (16)$$

- $\tau > 0$ gives a lower bound on $w_{EE}F'_E$:

$$w_{EE}F'_E > 1 + \frac{\tau_E}{\tau_I}(1 + w_{II}F'_I) \quad (17)$$

- $\Delta > 0$ gives an upper bound on $w_{EE}F'_E$:

$$w_{EE}F'_E < \frac{1 + w_{II}F'_I + w_{EI}w_{IE}F'_E F'_I}{1 + w_{II}F'_I} \quad (18)$$

4.2 Deriving the necessary condition

A feasibility window for $w_{EE}F'_E$ exists only when the lower bound (Equation 17) is less than the upper bound (Equation 18). Combining and simplifying:

$$w_{EI}w_{IE} > \frac{\tau_E}{\tau_I} \cdot \frac{(1 + w_{II}F'_I)^2}{F'_E F'_I} \quad (19)$$

Since F'_E and F'_I are not free parameters but determined by the fixed point, we minimise the right-hand side to obtain the weakest necessary condition.

Maximising F'_E . F'_E appears only in the denominator, so we need its maximum. From Equation 13, this is maximised when $F(x) + c = 1/2$, i.e. at the sigmoid midpoint $u_E = \theta_E$. Substituting:

$$F'_E{}^{\max} = a_E \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{a_E}{4} \quad (20)$$

Minimising over F'_I . Let $\varphi = F'_I > 0$. The right-hand side of Equation 19 becomes:

$$R(\varphi) = \frac{4\tau_E}{\tau_I a_E} \left(\frac{1}{\varphi} + 2w_{II} + w_{II}^2 \varphi \right) \quad (21)$$

Differentiating and setting to zero:

$$R'(\varphi) = \frac{4\tau_E}{\tau_I a_E} \left(-\frac{1}{\varphi^2} + w_{II}^2 \right) = 0 \quad (22)$$

$$\varphi^* = \frac{1}{w_{II}} \quad (23)$$

To confirm this is a minimum: $R''(\varphi) = \frac{4\tau_E}{\tau_I a_E} \cdot \frac{2}{\varphi^3} > 0$ for all $\varphi > 0$.

Substituting $F'_E = a_E/4$ and $F'_I = 1/w_{II}$:

$$R(\varphi^*) = \frac{4\tau_E}{\tau_I a_E} (w_{II} + 2w_{II} + w_{II}) = \frac{16\tau_E w_{II}}{\tau_I a_E} \quad (24)$$

This yields the necessary condition: if it fails, no fixed point can become unstable regardless of where it sits on the sigmoid.

$$w_{EI}w_{IE} > \frac{16\tau_E w_{II}}{\tau_I a_E} \quad (25)$$

Note: since F'_E and F'_I are jointly determined by the fixed-point coordinates, they cannot in general simultaneously attain their optimal values. This makes Equation 25 a potentially loose bound.

4.3 Verification

We verify Equation 25 against both parameter sets:

- **Li et al.:** $\frac{16 \times 20 \times 1}{10 \times 1} = 32$, and $w_{EI}w_{IE} = 520 \gg 32$. ✓
- **NMA:** $\frac{16 \times 1 \times 11}{2 \times 1.2} = 73.3$, and $w_{EI}w_{IE} = 52 < 73.3$. ✗

Note that for Li et al. parameters, $w_{II} = 1 < \frac{4}{a_I} = 4$, so the optimal $F'_I = \frac{1}{w_{II}} = 1$ exceeds the sigmoid maximum $\frac{a_I}{4} = 0.25$. The unconstrained bound gives 32; the constrained bound (using $F'_I = \frac{a_I}{4}$) gives $\frac{16 \times 20 \times (1 + 1 \times 0.25)^2}{10 \times 1 \times 1} = 50$. Since $520 \gg 50$, the conclusion is unchanged, but for parameter sets with $w_{EI}w_{IE}$ between 32 and 50, only the tighter bound is reliable.

4.4 Tightening via the I-nullcline

Equation 25 treats F'_E and F'_I as independent. In the actual system they are coupled through the fixed-point coordinates. We now exploit the I-nullcline geometry to reduce the two-dimensional optimisation to a single variable.

Key observation. The I-nullcline

$$r_I = F_I(w_{IE}r_E - w_{II}r_I + I_I^{\text{ext}}) \quad (26)$$

is independent of I_E^{ext} . Changing I_E^{ext} shifts the E-nullcline but leaves the I-nullcline unchanged, so the fixed point slides along the I-nullcline. Both sigmoid derivatives therefore become functions of a single coordinate:

$$\alpha(r_E) = F'_E(r_E), \quad \beta(r_E) = F'_I(r_I(r_E)) \quad (27)$$

and the right-hand side of Equation 19 becomes:

$$R(r_E) = \frac{\tau_E}{\tau_I} \cdot \frac{(1 + w_{II}\beta)^2}{\alpha \cdot \beta} \quad (28)$$

Logarithmic differentiation. To find the minimum of R , we differentiate $\ln R$:

$$\ln R = \text{const} + 2 \ln(1 + w_{II}\beta) - \ln \alpha - \ln \beta \quad (29)$$

$$\frac{d \ln R}{dr_E} = \frac{2w_{II}\beta'}{1 + w_{II}\beta} - \frac{\alpha'}{\alpha} - \frac{\beta'}{\beta} = 0 \quad (30)$$

where $\alpha' = d\alpha/dr_E$ and $\beta' = d\beta/dr_E$. Collecting the β' terms:

$$\beta' \cdot \frac{w_{II}\beta - 1}{\beta(1 + w_{II}\beta)} = \frac{\alpha'}{\alpha} \quad (31)$$

To evaluate β' , we differentiate the I-nullcline implicitly:

$$\frac{dr_I}{dr_E} = \frac{w_{IE}\beta}{1 + w_{II}\beta} \quad (32)$$

so that

$$\beta'(r_E) = a_I(1 - 2r_I - 2c_I) \cdot \frac{w_{IE}\beta}{1 + w_{II}\beta} \quad (33)$$

The decoupled optimum as a special case. When $\beta = 1/w_{II}$, the left-hand side of Equation 31 vanishes identically, forcing $\alpha' = 0$, i.e. $F'_E = a_E/4$. This is precisely the decoupled optimum used to derive Equation 25.

Explicit parameterisation. Using the second form (Equation 11) of the I-nullcline:

$$r_{E(r_I)} = \frac{1}{w_{IE}}(w_{II}r_I + F_I^{-1}(r_I) - I_I^{\text{ext}}) \quad (34)$$

both derivatives become explicit functions of r_I :

$$\beta(r_I) = a_I(r_I + c_I)(1 - r_I - c_I) \quad (35)$$

$$\alpha(r_I) = a_E(r_{E(r_I)} + c_E)(1 - r_{E(r_I)} - c_E) \quad (36)$$

so that $R(r_I)$ is fully explicit:

$$R(r_I) = \frac{\tau_E}{\tau_I} \cdot \frac{(1 + w_{II}\beta(r_I))^2}{\alpha(r_I) \cdot \beta(r_I)} \quad (37)$$

The I_E^{ext} that places a fixed point at $(r_{E(r_I)}, r_I)$ is obtained by inverting the E-nullcline:

$$I_E^{\text{ext}} = F_E^{-1}(r_E) - w_{EE}r_E + w_{EI}r_I \quad (38)$$

The minimum of $R(r_I)$ has no closed form but requires only a single numerical sweep over r_I with no root-finding.

Necessary condition. If $\min_{r_I} R(r_I) > w_{EI}w_{IE}$, then no I_E^{ext} can open the feasibility window at any point on the I-nullcline, and oscillation is impossible. This is strictly tighter than Equation 25.

5 Monte Carlo Validation

We sampled $N = 5000$ random parameter combinations ($w_{EE}, w_{EI}, w_{IE} \in [1, 30]$, $w_{II} \in [0, 20]$, $\tau_E, \tau_I \in [0.5, 30]$, $a_E, a_I \in [0.5, 3]$, $\theta_E \in [1, 10]$, $\theta_I \in [1, 20]$, $I_E^{\text{ext}}, I_I^{\text{ext}} \in [-2, 5]$) and compared the analytical predictions against numerical integration (RK4). To rigorously identify limit cycles, the initial transient period was discarded, requiring the subsequent trajectory to exhibit stable peak intervals (coefficient of variation < 0.1) and a non-trivial amplitude (> 0.01).

The analysis provides a concrete algorithm using the explicit parameterisation by r_I :

1. Compute $\min_{r_I} R(r_I)$. If $\min_{r_I} R(r_I) > w_{EI}w_{IE}$, oscillation is impossible for any I_E^{ext} (tighter necessary condition than Equation 25).
2. Identify the feasible interval: all r_I where $\tau > 0$ and $\Delta > 0$ (the instability region on the I-nullcline).
3. For each r_I in the instability region, invert the E-nullcline to obtain I_E^{ext} and count fixed points.
4. If *any* such I_E^{ext} yields a unique fixed point, the Poincaré-Bendixson theorem guarantees a limit cycle.

For each parameter set, an I-nullcline sweep selects the I_E^{ext} that maximises instability:

Stage	Count	Osc	Rate
Equation 25 fails	1698	0	0.0%
I-nullcline bound fails	2223	—	—
Bound passes, no unstable FP found	344	—	—
Unstable FP exists, multi-FP only	371	1	0.3%
Unique unstable FP found	364	356	97.8%

Table 3: I-nullcline sweep validation ($N = 5000$).

None of the 1698 samples that failed Equation 25 produced oscillation.

6 Limitations

Model scope. All results are derived for the simplified ODE formulation used in the NMA curriculum, which omits the refractory term $(1 - r_E)$. The original Wilson-Cowan (1972) ODE retains this factor; our condition would need to be rederived for that case.

Decoupled optimisation. Equation 25 is obtained by independently optimising F'_E and F'_I . In the actual system, both are jointly determined by a single fixed point and may not simultaneously attain their optimal values, making Equation 25 a potentially loose bound. When $w_{II} < 4/a_I$, the optimal $F'_I = 1/w_{II}$ exceeds the sigmoid maximum $a_I/4$, and the condition could be tightened.

Finite numerical scope. The Monte Carlo ($N = 5000$) is an empirical validation, not a proof.

7 References

1. Wilson, H.R. & Cowan, J.D. (1972). Excitatory and inhibitory interactions in localized populations of model neurons. *Biophysical Journal*, 12(1), 1–24.
2. Li, D., Liu, S. & Wang, J. (2022). Bidirectionally regulating gamma oscillations in Wilson-Cowan model by self-feedback loops. *Frontiers in Systems Neuroscience*, 16, 723237.
3. Neuromatch Academy — Computational Neuroscience, W2D4: Dynamic Networks.